

```
In [1]: # Support Vector Machine Classification

# Confusion Matrix

# Model report

# Evaluation Metrics= Accuracy, Specificity, Sensitivity, Precision, Recall,
f1 score, Macro Average, Weighted Average
```

```
In [2]: # importing libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [3]: dataset=pd.read_csv("Social_Network_Ads.csv")
```

```
In [4]: dataset
```

Out[4]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [5]: dataset=pd.get_dummies(dataset,drop_first=True)
```

```
In [6]: dataset
```

```
Out[6]:
```

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	1
1	15810944	35	20000	0	1
2	15668575	26	43000	0	0
3	15603246	27	57000	0	0
4	15804002	19	76000	0	1
...	...	...	...	...	...
395	15691863	46	41000	1	0
396	15706071	51	23000	1	1
397	15654296	50	20000	1	0
398	15755018	36	33000	0	1
399	15594041	49	36000	1	0

400 rows × 5 columns

```
In [7]: dataset["Purchased"].value_counts()
```

```
Out[7]: 0    257
        1    143
        Name: Purchased, dtype: int64
```

```
In [8]: indep=dataset[["Age","EstimatedSalary","Gender_Male"]]
        dep=dataset[["Purchased"]]
```

```
In [9]: indep.shape
```

```
Out[9]: (400, 3)
```

```
In [10]: dep
```

```
Out[10]:
```

	Purchased
0	0
1	0
2	0
3	0
4	0
...	...
395	1
396	1
397	1
398	0
399	1

400 rows × 1 columns

```
In [11]: # Splitting into training and test set
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test= train_test_split(indep,dep,test_size=0.3
0,random_state=0)
```

```
In [19]: from sklearn.svm import SVC
classifier= SVC(kernel='rbf', random_state=0)
classifier.fit(X_train,Y_train)
```

C:\Users\user\Anaconda3\lib\site-packages\sklearn\utils\validation.py:724: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

y = column_or_1d(y, warn=True)

C:\Users\user\Anaconda3\lib\site-packages\sklearn\svm\base.py:193: FutureWarning: The default value of gamma will change from 'auto' to 'scale' in version 0.22 to account better for unscaled features. Set gamma explicitly to 'auto' or 'scale' to avoid this warning.

"avoid this warning.", FutureWarning)

```
Out[19]: SVC(C=1.0, cache_size=200, class_weight=None, coef0=0.0,
decision_function_shape='ovr', degree=3, gamma='auto_deprecated',
kernel='rbf', max_iter=-1, probability=False, random_state=0,
shrinking=True, tol=0.001, verbose=False)
```

```
In [20]: Y_pred=classifier.predict(X_test)
```

```
In [21]: from sklearn.metrics import confusion_matrix
cm=confusion_matrix(Y_test,Y_pred)
```

```
In [22]: print(cm)
```

```
[[78  1]
 [33  8]]
```

```
In [23]: from sklearn.metrics import classification_report
clf_report=classification_report(Y_test,Y_pred)
```

```
In [24]: print(clf_report)
```

	precision	recall	f1-score	support
0	0.70	0.99	0.82	79
1	0.89	0.20	0.32	41
accuracy			0.72	120
macro avg	0.80	0.59	0.57	120
weighted avg	0.77	0.72	0.65	120

```
In [25]: print(cm)
```

```
[[78  1]
 [33  8]]
```

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In [ ]:
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In [ ]:
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